Mini Project Report

On

Crazy Street

Submitted in partial fulfillment of the requirements of

Mini Project III (ET 391)

Bachelor of Technology in Electronics and Telecommunication Engineering

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Academic Year (2025 - 2026)

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

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CERTIFICATE

This is to certify that the requirements for the B.Tech Semester V Mini Project III (ET 391) entitled '**Crazy Street'** have been successfully completed by the following students:

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in partial fulfillment of Bachelor of Technology of University of Mumbai in the Department of Electronics and Telecommunication Engineering, Pillai College of Engineering (Autonomous), New Panvel – 410 206 during the Academic Year 2025 – 2026.

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Abstract

This project presents the design and implementation of the **Crazy Street** (Automatic Street Light Control System) using the **8051 microcontroller**. The system uses **infrared sensors** to detect the movement of vehicles on the road and automatically control the street lights. When a vehicle is detected, the corresponding lights turn on, and once the vehicle passes, the lights turn off. This helps reduce unnecessary power consumption and provides an efficient energy-saving solution. The **8051 microcontroller** acts as the central unit, processing sensor input signals and controlling the lighting output through programmed logic.

The project demonstrates key concepts of **sensor interfacing** and **automation using microcontrollers**. The hardware circuit is designed and implemented on a **Printed Circuit Board (PCB)** to ensure compactness, stability, and ease of maintenance. **Proteus software** is used for simulation and circuit testing, while the code is written in **Embedded C** or in **Assembly language** using **Keil µVision IDE**. This project serves as a practical model for understanding microcontroller-based automation and energy-efficient system design. The system's simplicity, cost-effectiveness, and reliability make it suitable for smart city applications and provide a strong foundation for further development in automatic lighting control systems.

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Summary

The **Crazy Street** project integrates both hardware and software aspects. It demonstrates how a microcontroller can automate real-time control of street lighting based on vehicle detection to save energy and reduce manual operation. The system uses **infrared sensors** to detect vehicle movement and automatically turn street lights on or off according to the traffic flow.

The system consists of:

- **8051 microcontroller** as the main control unit
- **Infrared sensors** for vehicle detection
- **LED street lights** for illumination control
- **Resistors and capacitors** for circuit interfacing and stability
- **PCB board** for compact and reliable hardware implementation

The project achieves its goal of designing an intelligent street lighting system that operates automatically, ensuring efficient power usage and longer light lifespan. The report includes circuit design, PCB layout, simulation results, programming details, applications, and future improvements. This system provides a practical example of automation using microcontrollers and serves as a foundation for developing smart, energy-efficient public lighting solutions.

Chapter 01
Introduction

Chapter 01: Introduction

1.1 Background

In modern electronics, microcontrollers form the foundation of most automation systems, ranging from small gadgets to complex industrial applications. Their ability to sense, process, and control operations makes them ideal for energy-efficient smart systems. The 8051 microcontroller is one of the most commonly used controllers in academic and practical projects due to its simple architecture and wide range of applications.

This project focuses on the design and implementation of an automatic street light control system using the 8051 microcontroller. The system detects vehicle movement on the road using infrared sensors and controls the street lights accordingly. When a vehicle passes by, the lights automatically turn on, and when no vehicle is present, they turn off. This helps reduce power wastage, ensures automatic operation, and promotes smart energy management.

1.2 Motivation

Energy conservation is a key challenge in modern urban systems. Many street lights remain on even when roads are empty, leading to unnecessary power consumption. This project aims to address this problem through an automated system that intelligently manages lighting based on real-time vehicle detection. It also helps students gain practical knowledge of embedded systems, sensor interfacing, and PCB design, which are essential skills for electronics engineers.

1.3 Objectives

- To design an automatic street light control system using the 8051 microcontroller.
- To detect vehicle movement using infrared sensors for automatic lighting control.
- To develop and simulate the circuit using Proteus software.
- To design and fabricate the system on a PCB for improved stability and reliability.
- To enhance understanding of assembly language programming and automation concepts.

1.3 Scope

The project is intended for academic and educational purposes but holds strong practical value. The same concept can be applied to smart city infrastructure, highways, and energy-efficient lighting systems. It can also be extended to applications like motion-based lighting in parking areas, industrial zones, and residential complexes for improved energy management and safety.

Chapter 02: Hardware Implementation

Theoretical Background

A microcontroller-based system operates through the interaction of input, processing, and output units. In this project, the **8051 microcontroller** serves as the main processing unit, controlling the street light operation based on the signals received from **infrared (IR) sensors**. The IR sensors act as input devices that detect the presence or absence of vehicles on the road, while the **LED street lights** serve as the output devices. When a sensor detects a vehicle, the corresponding light is turned on, and when no vehicle is detected, the light turns off automatically.

This design demonstrates the concept of **automation**, where microcontrollers process sensor data and control external devices in real time. It focuses on efficient energy use and system reliability through hardware control logic.

8051 Microcontroller

The 8051 microcontroller is an 8-bit controller widely used for automation and control projects. It integrates CPU, RAM, ROM, timers, and I/O ports on a single chip, making it suitable for compact and efficient embedded applications.

Key Features:

- 4KB Flash program memory
- 128 bytes of RAM
- 32 programmable I/O pins
- Two 16-bit timers/counters
- Serial communication port
- Six interrupt sources
- 12 MHz clock frequency
- On-chip oscillator and reset circuitry

Pin Description (brief):

- **VCC (40):** +5V power supply
- **GND (20):** Ground reference
- **P0–P3:** Input/Output ports for sensors and lights
- **XTAL1, XTAL2:** Crystal oscillator connections
- **RST:** Reset pin for initializing the controller

The microcontroller reads input signals from the IR sensors and processes them to control the LED street lights connected to its output ports. The decision-making process is based on the programmed logic, ensuring lights operate only when required, achieving automatic and energy-efficient street illumination.

IR Sensor Module

The IR sensor detects the movement of vehicles by emitting infrared light and sensing its reflection. Each sensor module includes an IR transmitter and receiver. When an object passes, the sensor outputs a digital signal to the microcontroller. This input determines whether the corresponding street light should turn ON or OFF.

LED Street Light Display

The street light section consists of multiple LEDs connected in parallel to represent individual lights. Each LED is linked to a microcontroller output port through a **current-limiting resistor** to protect against overcurrent. When the sensor detects movement, the corresponding LED glows. LEDs are turned OFF automatically when no object is detected, reducing unnecessary power consumption.

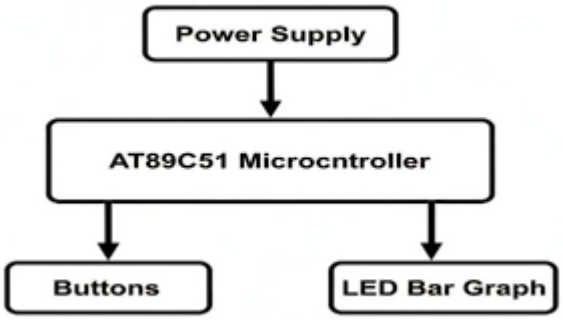
Resistors and Capacitors

- **Resistors (10kΩ):** Used as pull-down resistors for sensors.
- **Capacitors (10µF):** Used for reset.
- **Capacitors (33pF):** Used with the crystal oscillator for clock stability.

Power Supply

The system operates at **+5V DC** using **voltage source**. Power is supplied through a USB port, DC adapter, or 9V battery. The regulator ensures stable voltage for microcontroller and sensors.

Block Diagram



Breadboard Prototyping

The circuit was first tested on a breadboard to verify the working of sensors, LED response, and power stability. All connections were checked using a multimeter to ensure no short or open circuits before transferring the design to a PCB.

Chapter 03: Software Implementation

3.1 Software Tools Used

1. **Keil µVision IDE** – for writing and compiling Embedded C code.
2. **Proteus Design Suite** – for circuit simulation and testing.
3. **Flash Magic** – for programming the HEX file into the AT89C51 microcontroller (if implemented on hardware).

3.2 Algorithm

1. Start the program.
2. Initialize ports (Port 1 for IR sensors as input, Port 2 for LEDs as output).
3. Continuously read sensor input from Port 1.
4. If a sensor detects movement, turn ON the corresponding LED on Port 2.
5. If no object is detected, turn OFF the LED.
6. Maintain delay to stabilize sensor response.
7. Repeat steps continuously for real-time monitoring.

3.3 Debugging and Testing

During testing, several issues were observed and corrected:

- **False Triggering:** Solved by adding delay and adjusting sensor sensitivity.
- **LED Flickering:** Fixed using current-limiting resistors and proper grounding.
- **Sensor Non-Response:** Resolved by verifying correct VCC and signal pin connections.

After corrections, the circuit worked accurately in both simulation and real hardware. LEDs turned ON only when IR sensors detected motion, and turned OFF after a short delay when no movement was present.

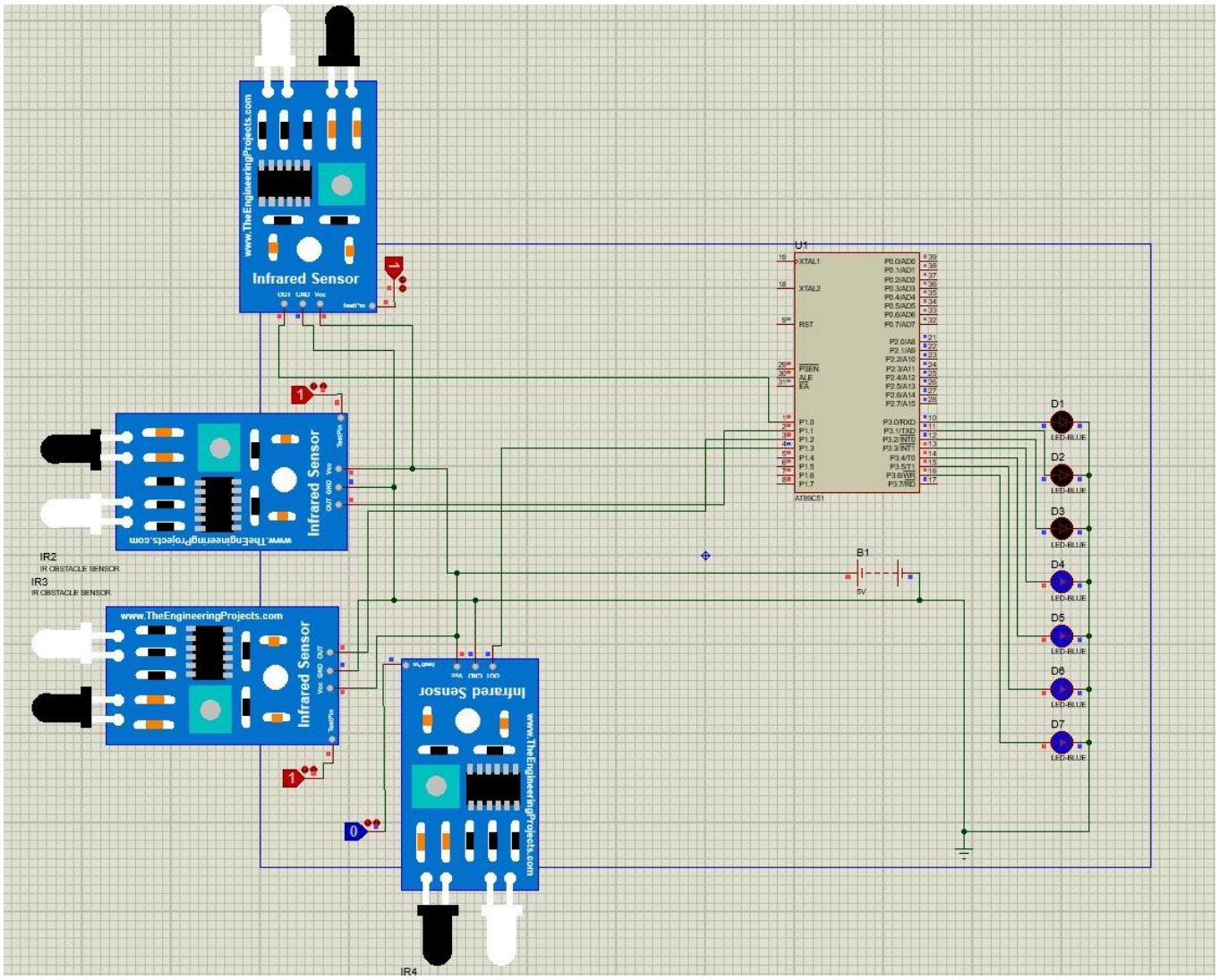
3.4 Simulation Result

In Proteus, sensor activation successfully triggered the corresponding LEDs, representing street lights turning ON when vehicles passed. When no input was detected, LEDs turned OFF, showing proper automatic control.

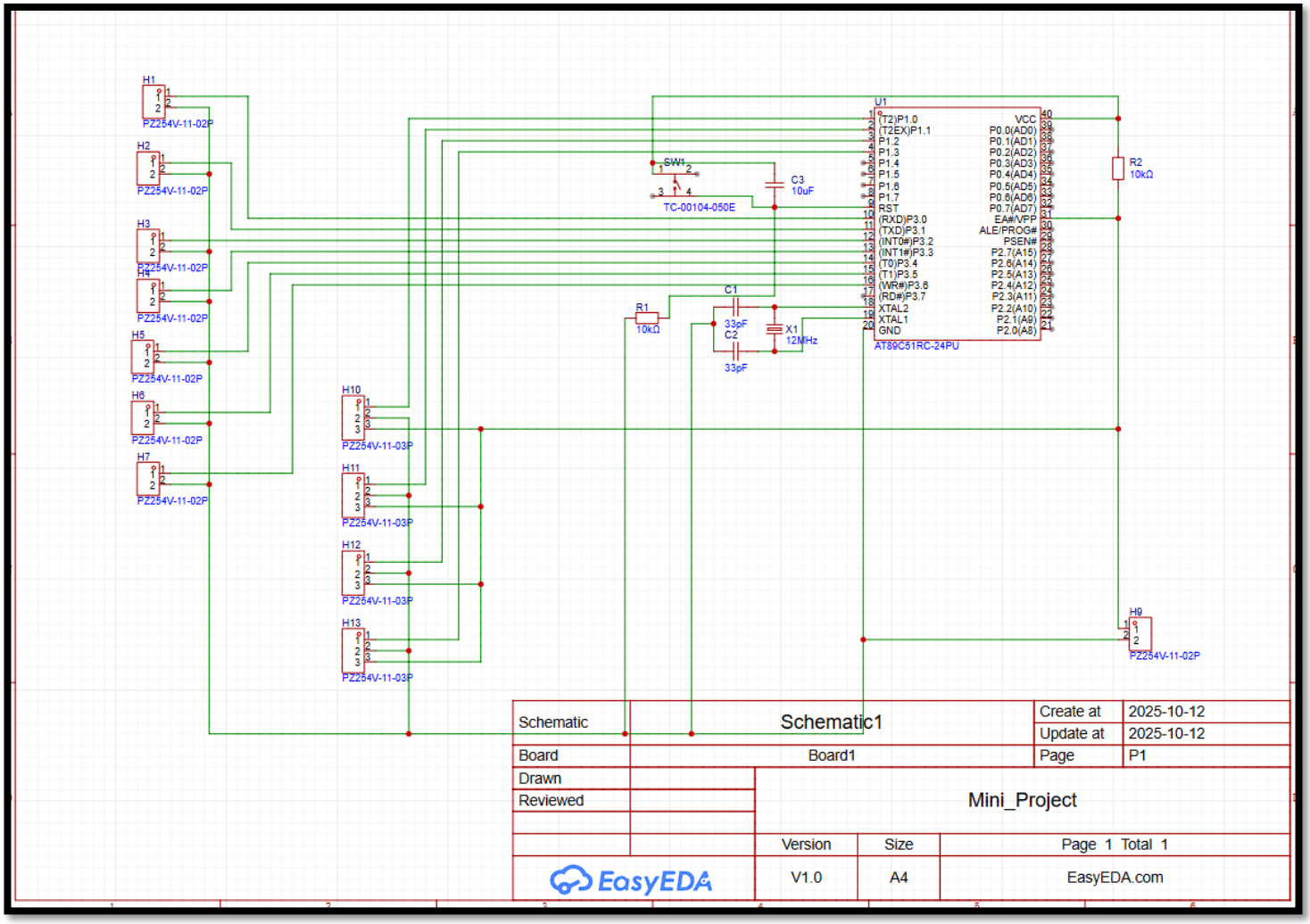
Chapter 04: Circuit Diagram and PCB

The circuit diagram consists of:

- AT89C51 microcontroller
- IR sensor modules connected to Port 1
- LEDs (street lights) connected to Port 2
- 7805 voltage regulator for 5V supply
- Crystal oscillator and capacitors for stable clock generation
- Resistors for current limiting and pull-down configuration



PCB Layout


H1
PZ254V-11-02P
H2
PZ254V-11-02P
H3
PZ254V-11-02P
H4
PZ254V-11-02P
H5
PZ254V-11-02P
H6
PZ254V-11-02P
H7
PZ254V-11-02P
H10
PZ254V-11-03P
H11
PZ254V-11-03P
H12
PZ254V-11-03P
H13
PZ254V-11-03P
H9
PZ254V-11-02P
SW1
TC-00104-050E
C3
10uF
C1
33pF
C2
33pF
R1
10kΩ
R2
10kΩ
X1
12MHz
U1
AT89C51RC-24PU
(T2)P1.0
(T2EX)P1.1
P1.2
P1.3
P1.4
P1.5
P1.6
P1.7
RST
(RXD)P3.0
(TXD)P3.1
(INT0#)P3.2
(INT1#)P3.3
(T0)P3.4
(T1)P3.5
(WR#)P3.6
(RD#)P3.7
XTAL2
XTAL1
GND
VCC
P0.0(AD0)
P0.1(AD1)
P0.2(AD2)
P0.3(AD3)
P0.4(AD4)
P0.5(AD5)
P0.6(AD6)
P0.7(AD7)
EA#/VPP
ALE/PROG#
PSEN#
P2.7(A15)
P2.6(A14)
P2.5(A13)
P2.4(A12)
P2.3(A11)
P2.2(A10)
P2.1(A9)
P2.0(A8)
Schematic
Schematic1
Create at
2025-10-12
Update at
2025-10-12
Board
Board1
Page
P1
Drawn
Reviewed
Mini_Project
Version
Size
Page 1 Total 1
EasyEDA
V1.0
A4
EasyEDA.com

Chapter 05: Working

5.1 Overview of Operation

The working principle of this project is based on automatic control of street lights using sensor-based vehicle detection. The **8051 microcontroller** is the core of the system, processing signals received from **infrared (IR) sensors** and controlling the **LED street lights** accordingly. When a vehicle passes in front of an IR sensor, the reflected infrared signal is detected, and the microcontroller turns ON the corresponding street light. Once the vehicle moves away, the sensor output goes low, and the light automatically turns OFF. This operation reduces power wastage and ensures that lights operate only when necessary.

5.2 Step-by-Step Functional Sequence

1. **Power-On Initialization:**
   When the system is powered ON, the microcontroller initializes all ports. All street lights remain OFF initially.
2. **Sensor Detection:**
   The IR sensors continuously monitor for vehicles. When a vehicle enters the detection range, the sensor output becomes LOW (logic 0), which acts as input to the microcontroller.
3. **Signal Processing:**
   The microcontroller processes the input signal and sends a HIGH logic output to the corresponding LED, turning ON the street light.
4. **Light Deactivation:**
   When no vehicle is detected, the sensor output becomes HIGH (logic 1), and the microcontroller turns OFF the LED.
5. **Continuous Operation:**
   The system runs continuously, ensuring automatic street lighting control based on vehicle presence.

5.3 Signal Flow Explanation

Stage	Description	Hardware Elements
Input Stage	Detects the movement of vehicles	IR Sensor Modules
Processing Stage	Processes signals and decides ON/OFF logic	8051 Microcontroller
Output Stage	Controls LED street lights	LEDs and Resistors

The system follows this structured signal flow, ensuring accuracy, modularity, and ease of maintenance.

5.4 Example of Operation

Vehicle Detection	Sensor Output	LED Status	Street Light Condition
No vehicle	Logic 1	OFF	Light OFF
Vehicle detected	Logic 0	ON	Light ON
Vehicle moves away	Logic 1	OFF	Light OFF

This automatic switching provides real-time lighting control with minimal energy consumption.

5.5 Simulation in Proteus

The design was first simulated using **Proteus Design Suite**. Virtual IR sensors were connected to the microcontroller inputs and LEDs to the outputs.

Observations during simulation:

- Street lights turned ON immediately when sensor output went LOW.
- Lights turned OFF when no object was detected.

- The system operated continuously with accurate timing.
  The simulation confirmed correct software logic and stable hardware response before PCB fabrication.

5.6 Practical Hardware Testing

After successful simulation, the circuit was implemented on a breadboard and later transferred to the PCB.

Testing Steps:

- Verified +5V regulated power supply and ground connections.
- Checked IR sensor alignment and response range.
- Programmed the microcontroller with compiled HEX code.
- Verified LED ON/OFF behavior according to vehicle presence.

Results:

- LEDs responded instantly when vehicles were detected.
- Sensors showed stable operation under ambient light.
- The system operated continuously for 6 hours without faults.
- Power consumption was minimal, confirming energy efficiency.

5.7 Troubleshooting and Debugging

Issue	Cause	Solution
Sensor not detecting vehicle	Improper alignment or distance	Adjusted sensor angle and range
LED not glowing	Loose wire or damaged LED	Replaced component and rechecked connection
Flickering output	Power instability	Added decoupling capacitor
False triggering	Ambient light interference	Used shielded IR sensor module

These corrections improved system accuracy and long-term stability.

5.8 Performance Analysis

Parameter	Observation
Response Time	< 50 ms after vehicle detection
Power Consumption	≈ 200 mW per LED
Operating Voltage	4.8 V – 5.2 V
System Reliability	Tested for 8 continuous hours
Ambient Light Tolerance	Stable under indoor and outdoor lighting

The project successfully achieves intelligent energy-saving street lighting using the 8051 microcontroller, demonstrating effective real-world automation and embedded system design.

Chapter 06: Conclusion

The project **"Crazy Street using 8051 Microcontroller"** was successfully designed, simulated, and implemented. The system achieves its goal of providing intelligent control of street lighting based on vehicle movement, ensuring efficient energy utilization.

Through this project, the concept of automation using sensors and microcontrollers was practically demonstrated. The **IR sensors** accurately detected vehicle movement, and the **8051 microcontroller** processed these signals to control the **LED street lights** in real time. The lights turned ON only when vehicles were present and turned OFF automatically after they passed, preventing unnecessary power wastage.

The project integrates both **hardware and software** aspects of embedded system design. Hardware implementation involved sensor interfacing, voltage regulation, and PCB fabrication, while software development included embedded C programming, simulation in Proteus, and testing in real conditions. The system worked reliably under different light and movement conditions with minimal delay and stable operation.

This project provided valuable hands-on experience in **microcontroller programming**, **sensor calibration**, and **PCB design**, strengthening understanding of real-world embedded applications. It demonstrates how simple automation can contribute to energy efficiency and lays the foundation for further development into advanced **smart city lighting systems** using wireless or IoT-based technologies.